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3 April 1962
SUPERSEDING
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10 May 1957

MILITARY SPECIFICATION

PLASTIC MATERIAL, CELLULAR, POLYSTYRENE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.- This specification covers cellular expanded polystyrene intended primarily for thermal insulation and buoyancy applications.

1.2 Classification.- Cellular expanded polystyrene shall be of the following types, classes and grades, as specified (see 6.2):

Type I - Thermal insulation.
  Class 1 - Non-fire retardant.
  Class 2 - Fire retardant.
  Grade A - 8 p.s.i. minimum compressive strength.
  Grade B - 15 p.s.i. minimum compressive strength.
Type II - Buoyant.
  Class 1 - Non-fire retardant.
  Class 2 - Fire retardant.
  Grade A - Low density.
  Grade B - High density.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
C-C-91 - Candles, Illuminating.
PPP-B-636 - Box, Fiberboard.

MILITARY
MIL-F-859 - Fuel Oil, Burner.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS
C177-45 - Thermal Conductivity of Materials by Means of the Guarded Hot Plate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Materials.- Cellular polystyrene shall consist of polymerized styrene expanded many times in volume. It may be formed by the expansion of high density beads or granules in a mold or directly from the base resin by extrusion. The material shall be firm in composition and essentially unicellular.

3.2 Uniformity.- The material shall be homogeneous and of uniform texture and density. There shall be no visible large void openings, accumulations of unexpanded resin, inclusions, tears, or tears, that have been cemented together. Surfaces shall be smooth, edges and corners well defined, and there shall be no areas containing a filled-in material.

3.3 Color.- The color of class 1 material shall be white or the natural color of the product. Class 2 material shall be colored or suitably marked to distinguish it from class 1 material.

3.4 Odor.- The material shall be free from any objectionable odor.

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3.5 Unless otherwise specified (see 6.2 and 6.3), board form shall be furnished.

3.6 Edges. - The edges of board form cellular polystyrene shall be square. Rough planks and billets for buoyancy may have rounded edges as manufactured with skin.

3.7 Dimensions. - The dimensions of the material shall be as specified (see 6.2 and 6.3).

3.8 Dimensional tolerances. -

3.8.1 Length. - Unless otherwise specified (see 6.2), for lengths up to 48 inches, the tolerance shall be ±1/4 inch. For greater lengths, the tolerance shall be ±2 percent, -0.50 percent.

3.8.2 Width. - Unless otherwise specified (see 6.2), the width shall have a tolerance of ± 1/8 inch.

3.8.3 Thickness. - Unless otherwise specified (see 6.2), the thickness shall have a tolerance of ± 1/16 inch.

3.9 Type I, (grades A and B), thermal insulation. -

3.9.1 Density. - The maximum density shall be 2.8 pounds per cubic foot when tested as specified in 4.5.5.

3.9.2 Water absorption. - The maximum water absorption shall be 0.12 pounds per square foot of skinless or rindless surface when tested as specified in 4.5.7.

3.9.3 Thermal conductivity. - The thermal conductivity (K value - B. T. U. per hour per square foot of insulation for 1°F. gradient per inch thickness) shall not exceed 0.28 and 0.26 at the mean temperatures of 75°F. and 40°F., respectively when tested as specified in 4.5.12.

3.9.4 Dimensional stability. - The specimen shall not change more than 0.5 percent in a linear direction when tested as specified in 4.5.8.

3.9.5 Compressive strength. -

3.9.5.1 Grade A. - The minimum compressive strength at not more than 5 percent deflection shall be 8 p.s.i. when tested as specified in 4.5.6.

3.9.5.2 Grade B. - The minimum compressive strength at not more than 5 percent deflection shall be 15 p.s.i. when tested as specified in 4.5.6.

3.9.6 Fire resistance (class 2 only). - The material shall be self extinguishing as defined in 4.5.11.2 and when tested as specified in 4.5.11.

3.10 Type II, (grades A and B), buoyant. -

3.10.1 Water absorption. - The maximum water absorption shall be 0.12 pounds per square foot of skinless or rindless surface when tested as specified in 4.5.7.

3.10.2 Volume change on heat aging. - The specimen shall not change more than 3.5 percent in volume when tested as specified in 4.5.9.

3.10.3 Oil resistance. - The material shall show no apparent softening or swelling when tested as specified in 4.5.10.

3.10.4 Fire resistance (class 2 only). - The material shall be self extinguishing as defined in 4.5.11.2 and when tested as specified in 4.5.11.

3.10.5 Density. -

3.10.5.1 Grade A. - The maximum density shall be 2.5 pounds per cubic foot when tested as specified in 4.5.5.

3.10.5.2 Grade B. - The maximum density shall be 6.0 pounds per cubic foot when tested as specified in 4.5.5.

3.10.6 Compressive strength. -

3.10.6.1 Grade A. - The minimum compressive strength at not more than 5 percent deflection shall be 5.0 p.s.i. when tested as specified in 4.5.6.

3.10.6.2 Grade B. - The minimum compressive strength at not more than 5 percent deflection shall be 100.0 p.s.i. when tested as specified in 4.5.6.

3.11 Workmanship. - The workmanship shall be the best commercial practice covering this type material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Inspection lot. - For purposes of sampling, a lot shall consist of all material of one type, class

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and grade manufactured at the same time, in one production run, and offered for delivery at the same time.

4.2.2 Unit. - A unit shall consist of a single board, plank, or billet of unpackaged material.

4.2.3 Sampling for examination. - A random sample of material units (see 4.2.2) shall be selected from each lot in accordance with Standard MIL-STD-105 at inspection level II. The acceptable quality level shall be equal to 2.5 percent defective. Samples shall be inspected in accordance with 4.3.

4.2.4 Sampling for tests. - A random sample of material units (see 4.2.2) shall be selected from each inspection lot in accordance with Standard MIL-STD-105 at inspection level L-8 for group A tests, level L-5 for group B tests and L-1 for group C tests. Specimens from each sample unit shall be subjected to the applicable tests specified in 4.5. If any specimen fails in any test, this shall be cause for rejection of the lot represented by the sample.

4.2.4.1 Group A tests. - Group A tests shall include:

(a) Density (see 4.5.5).
(b) Compressive strength (see 4.5.6).
(c) Fire resistance (class 2 only) (see 4.5.11).

4.2.4.2 Group B tests. - Specimens for group B tests may be obtained from the samples selected for group A tests. Group B tests shall include:

(a) Water absorption (see 4.5.7).
(b) Volume change on heat aging (type II only) (see 4.5.9).
(c) Dimensional stability (type I only) (see 4.5.8).
(d) Oil resistance (type II only) (see 4.5.10).

4.2.4.3 Group C tests. - Sets of specimens for group C tests may be obtained from samples selected for group A tests. Group C tests shall include:

(a) Thermal conductivity (type I only) (see 4.5.12).

4.3 Examination. - Each of the sample material units selected in accordance with 4.2.3 shall be visually, dimensionally, and odor examined to verify compliance with this specification. When measuring the dimensions of units for inspection, any convenient measuring rule graduated in 1/16 inch shall be used. Any material unit in the sample containing one or more visual, dimensional, or odor defects, shall not be offered for delivery. If the number of defective material units in any sample exceeds the acceptance number for the sample, this shall be cause for rejection of the lot represented by the sample.

4.4. Inspection reports. - Three copies of inspection reports shall be forwarded to the bureau or agency concerned, via the inspector.

4.5 Test procedures. -

4.5.1 Conditioning of test specimens. - Unless otherwise specified, all tests shall be conducted at a temperature of 23° + 1° Centigrade (C) (73.4±1.8° F.) on specimens conditioned for at least 5 hours in the same atmosphere prior to testing.

4.5.2 Test specimen selection. - From each sample unit selected in accordance with 4.2.4, one specimen shall be cut for each test specified in 4.5 except for the thermal conductivity test of 4.5.12. For thermal conductivity test one set of 2 specimens, shall be cut from the sample unit.

4.5.3 Dimensional measurements. -

4.5.3.1 Thickness. - Unless otherwise specified, test specimens shall be measured to the nearest 0.001 inch in at least three places with a dial guage micrometer or equivalent and the thickness taken as as average of these three readings. Unless otherwise specified, the micrometer shall have a flat presser foot with an area of one square inch; the other flat bearing surface shall not be smaller than the presser foot and the presser foot shall exert a force of four ounces. All measuring devices shall be graduated to read to 0.001 inch.

4.5.3.2 Linear dimensions. - Except where otherwise specified, length, width or diameter of the test specimen shall be measured to the nearest 0.01 inch. Each dimension shall be measured at three equally spaced places and the dimension taken as the average of these three readings. All measuring devices shall be graduated to read to 0.01 inch.

4.5.4 Weight. - Unless otherwise specified, the weight of test specimens shall be taken to the nearest 0.1 gram.

4.5.5 Density. - The density of the material shall be determined by dividing the weight of the material by its volume, and shall be expressed in pounds per cubic foot. The volume shall be determined by measuring the dimensions of a specimen approximately 1 foot square in cross section by its thickness. If thickness is greater than 1 inch, it may be measured to the nearest 0.01 inch. If thickness of the specimen is greater than 6 inches, cut down to a 6-inch thickness. Density, expressed in pounds per cubic foot $= \frac{\text{weight (grams)}}{\text{volume (cubic inches)}} \times 3.81$.

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4.5.6 Compressive strength. -

4.5.6.1 Specimens. - Test specimens shall be 6 by 6 inches square and approximately 1 inch in thickness.

4.5.6.2 Procedure. - The test specimen shall be placed between the platens of a standard compression testing machine and subjected to a gradually increasing load (speed of test platen should be 0.1 inch per minute) applied uniformly to the faces of the material to the point of failure indicated by crush or break, or until specimen is compressed 5 percent of original thickness, whichever occurs first. The maximum load in pounds per square inch (p.s.i.) and percent compression at that load shall be noted.

4.5.7 Water absorption. -

4.5.7.1 Specimens. - Test specimens shall be 4 by 4 inches square and approximately 1 inch in thickness. Any skin or smooth surface on the specimen should be removed by cutting or sawing so as to provide a natural or cellular surface.

4.5.7.2 Procedure. - The specimens shall be weighed and submerged in water under a 10-foot head of water (equal to 4.3 p.s.i.) at room temperature (65-80°F.) for 48 hours. The specimens shall then be placed in a stream of compressed air at a line pressure of 80 p.s.i. to blow the surface water from the specimen. A nozzle shall be used to direct the air at the specimen which shall be held in one hand with a four by four inch side up. The nozzle shall be held about 2 inches from the surface of the specimen and at an angle of about 30°. Starting at one corner, the air shall be played back and forth across the width of the specimen until the entire surface has been covered. Each of the other five surfaces shall be blown off in a like manner. This procedure shall be repeated until all of the surface water has been removed. The specimen shall then be reweighed. The results shall be calculated in terms of pounds per square foot of rindless or skinless surface.

4.5.8 Dimensional stability (type I only). - Specimens from each of the test units shall be cut 12 by 3 by 1 inch. At each of two points approximately 10 inches apart on the center line of the specimen place a bench mark[1/]. The specimens shall be conditioned for 24 hours at a temperature of 23°±1°C. and measurements made between the bench marks with a scale graduated to 0.01 inch to the nearest 0.01 inch. The specimen shall be placed flatwise on an expanded metal shelf in an air circulating oven equipped with a temperature control to maintain a 158 ± 2°F. temperature during the test. After 48 hours the samples shall be removed from the oven and conditioned for at least 2 hours at 23±1°C. The dimensional stability shall be reported as the change in length between bench marks expressed as a percentage of the length measured initially.

4.5.9 Volume change on heat aging (type II only only). -

4.5.9.1 Specimens. - Test specimens shall consist of squares 4 inches by 4 inches by 1 inch (approximately) in thickness.

4.5.9.2 Procedures. -

4.5.9.2.1 Volume before and after the heat aging test shall be determined by measuring the length, width and thickness of the specimen and calculating the volume from these measurements (see 4.5.3).

4.5.9.2.2 The specimens shall be placed in an oven maintained at 140°±2°F. for a period of 48 hours. At the end of that period the specimens shall be removed from the oven and allowed to recover for 5 hours at 23 ± 1°C. before the measurement of final volume is made. The volume change shall be reported as a percentage of the volume measured initially.

4.5.10 Oil resistance (type II only). -

4.5.10.1 Specimens. - The test specimen shall be a disk approximately 1.129 inches in diameter and 1 inch (approximately) in thickness.

4.5.10.2 Procedure. - The specimen shall be immersed in fuel oil conforming to grade Navy special of Specification MIL-F-859 for 70 hours. The specimen shall then be removed and blotted with filter paper. The specimen shall then be compared to an untreated specimen of similar size for apparent softness and visible swelling.

4.5.11 Fire resistance (class 2 only). -

4.5.11.1 Specimens. - The test specimens shall be 1/4 inch in thickness, 1 inch in width and a suitable length[2/]. Thickness and width tolerance shall be plus or minus 1/32 inch.

4.5.11.2 Procedure. - The test shall be performed in a location free from drafts. The specimen shall be clamped at one end in a position such that the long (length) dimension forms a 45-degree angle down from the horizontal and with the width in a vertical position. A candle flame shall be applied

1/ A suitable method of bench marking is to use staples.

2/ The length shall be sufficient for the ignition period and maximum burning time.

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to the lower or free end of the specimen for 15 seconds. The candle shall then be removed. If the specimen melts or shrinks from the flame, during the ignition period, the candle flame shall continue to be applied to the remaining end of the molten or shrunken specimen. The specimen shall be rated self extinguishing if there is no flaming or burning 10 seconds after removal of the candle flame. The candle used shall conform to class B of Specification C-C-91.

4.5.12 Thermal conductivity (type I only). - Thermal conductivity shall be determined by the guarded hot plate method specified in ASTM Specification C177-45 with the material dried in a dessicator at from 120-140°F. until excess moisture is driven off as indicated by a constant weight determination before testing.

4.6 Inspection of preparation for delivery. - The packing and marking of the material shall be examined to determine conformance with the requirements of this specification.

4.6.1 Examination for conformance to the packing and marking requirements not covered by referenced specifications shall be in accordance with Standard MIL-STD-105 using AQL of 4.0 percent defective at inspection level L-3.

5. PREPARATION FOR DELIVERY

5.1 Packing. - Packing shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. - Cellular polystyrene (boards and planks) shall be packed in fiberboard boxes conforming to class 2 of Specification PPP-B-636 except that dimensional limitations shall be waived.

5.1.2 Level B. - Cellular polystyrene (boards and planks) shall be packed in fiberboard boxes conforming to class 1 of Specification PPP-B-636 except that dimensional limitations shall be waived.

5.1.3 Level C. - The plastic material shall be packed in containers in a manner to insure acceptance by common carrier and will afford protection against physical and mechanical damage during shipment from the supply source to the first receiving activity, for immediate use. Shipping containers shall comply to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.2 Marking. - In addition to any special marking specified in the contract or order or herein, shipping containers shall be marked in accordance with Standard MIL-STD-129. Each pack shall be marked indicating type, class, and grade.

6. NOTES

6.1 Intended use. - Cellular polystyrene is intended for low temperature (down to - 100°F.) insulation and buoyancy applications. The maximum recommended continuous use and storage temperature is ± 158°F.

6.1.1 Type I material. - Type I, grade A material is intended for thermal insulation application where a compressive strength of 8 p.s.i. is satisfactory; type I, grade B material can be used where either 8 p.s.i. or 15 p.s.i. compressive strength is desired. Higher compressive strength material for use as thermal insulation is available.

6.1.2 Type material. - Type II, grade A material is intended for buoyancy applications where a compressive strength of 5 p.s.i. is satisfactory. This material with higher compressive strengths is available. Type II, grade B material should be used where high compressive strengths are required for buoyancy and also could be used for thermal insulation requiring high compressive strengths provided the thermal conductivity and dimensional stability characteristics are satisfactory.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and grade required (see 1.2).
(c) Other forms (rough planks and billets) required (see 3.5 and 6.3).
(d) Dimensions required (see 3.7 and 6.3).
(e) Other dimensional tolerances, if any (see 3.8).
(f) Level of packing required (see 5.1).

6.3 Sizes. - Cellular polystyrene is commercially available in the following board sizes: 18 by 108 inches, 18 by 36 inches and 24 by 48 inches with the thickness of 1, 1-1/2, 2, 3, 4 and 6 inches. Other forms such as rough planks and billets and other sizes in widths up to 48 inches and lengths up to 144 inches are available.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder, or any

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Custodians:
Army - SigC
Navy - Ships
Air Force - ARDC

Preparing activity:
Navy - Ships
(Project 9330-0068)

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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